

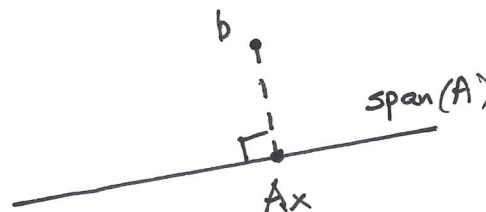
AM 205 Quiz 1, 7 October 2024

No books, no notes, no phones, no computers. Do five of the six problems; all count equally. If you answer six, your score will be based on the best five. The quiz is allotted 60 minutes, but you may need less time.

In each case give a sentence or two of clear explanation, with a sketch if you wish. None of these answers needs to be lengthy, but you must write a few words so that we can follow your reasoning.

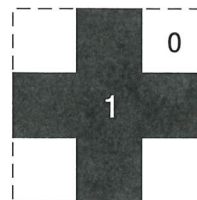
1. In standard IEEE double precision arithmetic, which is the smallest positive integer that is not a floating point number?

2. On the right is a figure we have drawn many times to illustrate least-squares fitting, i.e., approximation in the 2-norm. With $\text{span}(A)$ and b remaining in the same places, draw the corresponding figure for approximation in the ∞ -norm. There is an angle involved in this sketch: make sure you specify exactly what this angle is.



3. In lectures we talked about Gaussian elimination for a matrix A both in its classical (“20th century”) role and in another newer (“21st century”) role. In the classical context, people liked A to be well-conditioned: write one sentence explaining why. In the newer context, people like A to be ill-conditioned: write one sentence explaining why.

4. What’s the rank r of this “Swiss flag” image? Your answer should state the precise number r and show that the rank is neither $> r$ nor $< r$.



5. Let A be an $n \times n$ dense matrix and B a $2n \times n$ dense matrix, and suppose we compute LU factorizations by Gaussian elimination with partial pivoting (or QR factorizations by the Householder method; the answers are the same). In the limit $n \rightarrow \infty$, what’s the ratio of the number of floating point operations for factoring B to the number for factoring A ?

6. Let A and B be two $n \times n$ matrices with the property that $\|AB\| = \|A\|\|B\|$ (with $\|\cdot\| = \|\cdot\|_2$). What does this tell us about singular vectors of A and B ?